

A Comparative Study on Power Consumption in Smartphone technology through communication standards

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ABSTRACT- In twenty first centuries demand for smart phones and mobile primarily based applications is enlarged suddenly from past few years .now a day's smart phones acquire the place of standard computers and laptops up to some limit. Due to smaller, lighter and simple to relinquishment. However, we've some limit of power in Smartphone technology. The new generation of communication commonplace system like third generation(3G),4th generation(4G),5th generation(5G), will provide higher service and performance to Smartphone users, this generation consumes additional power in smart phones .the term of battery backup is refer as electric battery life. This survey paper presents varied concepts and proposals for reducing the facility consumption in Smartphone technology which provides additional support to the new generation of communications.

Keywords: Smartphone Technology, Communication Standard, Ultra FDD architecture, power consumption, wireless technology evolution.

I. INTRODUCTION

Smartphone needed some energy for his or her performing from storage devices. Except for some energy storage devices like smart phones battery restricted due to size and weight of the device. This means that energy potency and performance of those devices is extremely crucial to their usability. Hence, best management of power consumption of those devices is crucial. Most of the electronic widget is power-driven by storage batteries. For the designer and engineer square measure to provide the most effective device within the market with improved battery backup, performance, and life. A Smartphone may be a excellent example of the electronic devices and is charming several engineers and researchers during this field. The communication standards like 3G, 4G, and 5G need giant power for his or her process. They're facing battery backup issues in Smartphone. High rate services like audio/video streaming /gaming conference, video on demand, quicker web and plenty of alternative things beside the fundamental audio decision service. The mobile makers square measure competitor with one another to provide Smartphone with improved battery life and performance. It's imperative to optimize and management energy usage with efficiency in network parts. The hardware parts of a typical Smartphone square measure processor/CPU, RAM (Random Access Memory), LCD, screen, electronic equipment (2G/3G/4G/5G all of it), GPS, Wi-Fi, Bluetooth, Hot-spot, NFC etc. The economical energy management needs a decent understanding of wherever and the way the energy is optimized. The paper gift a close analysis of the facility consumption of Smartphone, the Openmoko modern Freerunner . in Paper, measurements overall system power, yet as activity of power consumption in main hardware parts of device .new generation communication technologies, provides high-speed rate services. this can be thanks to the upper information measure allotted to those technologies. Thence power drain area

Additional comparison 2G. To amend battery life and performance, it's necessary to effectively optimize power consumption in each attainable use case situation. With such optimization technique, one additionally ought to things is that the performance of the general system isn't affected, means that the output and quality of service (QoS) mustn't be affected. The rest of this paper is organized as follows. Section 2. The related work in power optimization techniques in smartphones. Section 3. Discusses the methodology along with the previous background study. Section 4. Discusses about power consumption analysis in smartphones technology and Section 5. The wireless technology evolution 1G, 2G, 3G and 4G. Section 6. Conclusion

II. RELATED WORK

Experiment performs a power consumption analysis of on a laptop system. Approach to component power measurement is driven partially by direct power measurement, but largely by deduction using modeling and off-line piece-wise analysis. It show that the display and CPU are the main consumers of energy for their class of system, and that other components contribute substantially only when they are used intensively. Their results same like our observations that RAM power is insignificant in workloads. The analysis of power consumption in a smartphone various usage scenarios is described by 'Openmoko Neo Freerunner' for 2G (GSM) connections by developing a power models [1]. In [2] the paper experiment identifies many components during various states of the smart phone where the power consumption will be more and critical. An comparison and analysis of power consumption measurements in smart phones for 2G, 3G networks is presented for the services-text messaging, voice and data.[3]The authors give the details information about evolution of wireless technology from 1G to 5G and it shows some specifications which is very helpful for our survey.[4] power consumption in Mobile devices they performed two experiments, first measures consumption for a set of usage scenarios, which represent common daily user activities, while second one analyzes a context-aware application with source code.[5] The battery powering a smartphone also has a size constraint and hence the capacity. Though communication standards systems such as 3rd Generation (3G), 4th Generation (4G) and 5th Generation (5G) can provide the high speed services to users, they draw considerably large power in smartphones. This limits the battery backup of smart phones and hence the battery life. This survey presents various ideas and proposals published for reducing the power consumption in smartphones which support new generation communication standards. [6] Thesis is presented on Radio Resource management in 3G UMTS Networks which is used in 3G Smartphone technology. In [7] authors propose an evolution of these algorithms in the forward link. In [9] context –aware battery management for mobile phones presents the information about computing and communication generations. In [10] paper presents on evolution of all communication generations which gives technical specifications which is very helpful for any further study.

UMTS (Universal Mobile Communication Systems) is a 3rd generation (3G) telecommunication network based on the existing GSM (2nd generation, 2G) core network employing new radio access technology WCDMA (Wideband Code Division Multiple Access). LTE (Long Term Evolution) is a 4G speed is (200Mbps to 1Gbps) it gives high end live video streaming but 3G was developed mainly to provide high speed data rate (up to 2 Mbps) services like internet, video telephony, multimedia and audio/video streaming which 2G could not support due to bandwidth limitations. But 4G technologies give high end live video streaming. From 4G we fill the live streaming video calls.

III.METHODOLOGY

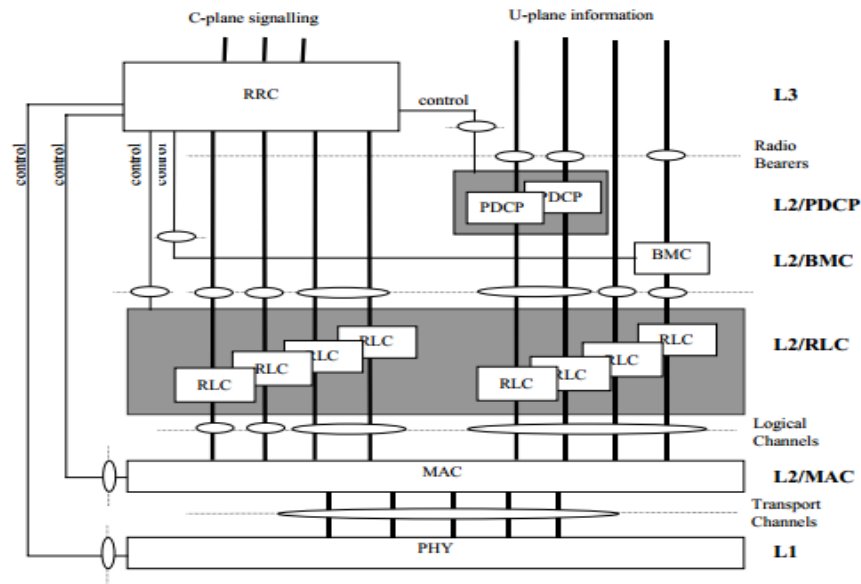


Figure 1. UTRA FDD Radio Interface protocol architecture.

The UMTS -Universal Mobile Telecommunication System is a third generation mobile communication system, designed to support a wide range of applications with good quality of service (QoS). The 3G has efficiency of transporting wideband and high bit rate multimedia services with some traditional cellular services e.g. voice, messaging etc. To provide these services with better service quality and enhance the performance of wireless network, management of radio resources is necessary. To do this, UMTS tender many radio resource management (RRM) techniques. These RRM providing different services with better quality with keeping the end user happy and stable network. In paper, main objective is to explore some RRM techniques and understand their practical importance with simulation of RRM algorithms. Starting with UMTS overview and some important concept about UMTS architecture. Then going deep into physical layer of UMTS. In the wake of getting solid idea of UMTS radio design and methodology, chipped away at various RRM procedures and at last, investigate two power control calculations to comprehend and get some pragmatic experience of genuine RRM systems, in light of the fact that power control is the vital most and basic piece of RRM methods due to interference limited nature of CDMA systems. WCDMA is a wideband Direct-Sequence Code Division Multiple Access (DS-CDMA) system. Portable User data bits are spread over a wide transfer speed by duplicating the client information with semi arbitrary bits (called chips). The chip rate of 3.84 Mcps (Million chips for every second) prompts a bearer transfer speed of around 5 MHz. It bolsters very factor client information rates, as such the idea of getting Bandwidth capacity on Demand (BoD) is all around bolstered. It underpins two fundamental methods of operation-Frequency Division Duplex (FDD) and Time Division Duplex (TDD). In the FDD mode, isolate 5 MHz transporter frequencies are utilized for the uplink and downlink separately, while in TDD just a single 5 MHz is timeshared between the uplink and downlink. Uplink is the association from the versatile to the base station, and downlink is that from the base station to the portable. The radio interface traditions are required to set up, reconfigure and release the Radio Bearer organizations. The convention engineering [9] is appeared in Figure 1. The convention layers over the physical layer are known as the information interface layer (Layer2) and the system (Layer 3). In the UTRA FDD radio interface, Layer 2 is part into sub layers. In the control plane, Layer 2 contains two sub-layers – Medium Access Control

(MAC) convention and Radio Link Control (RLC) convention. In the client plane, notwithstanding MAC and RLC, two extra administration subordinate conventions exist – Packet Data Convergence Protocol (PDCP) and Broadcast/Multicast Control Protocol (BMC). Layer 3 comprises of one convention, called Radio Resource Control (RRC) that incorporates a place with the management plane. There are 2 interfacing areas within the UMTS network, one is that the infrastructure domain that consists of the core network (CN) and second is that the UTRA (UTRAN) network, additionally user equipment (UE) domain. The UTRAN that consists of mobile station, the bottom station (Antenna, transceiver and controller) and radio interface is there between mobile station & base station. In UMTS network, the core network that's referred to as CN has the most responsibility to supply change and routing for user traffic.

IV. POWER CONSUMPTION ANALYSIS

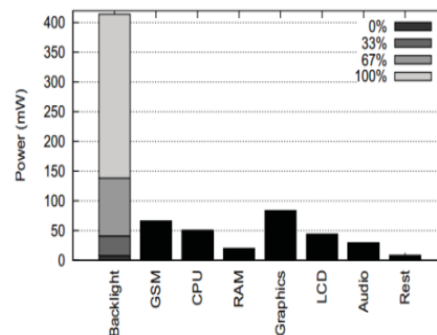


Figure 2 . Power breakdown for sending an SMS

Power consumed for causing associate degree SMS is dominated by the show elements and is as shown in Figure. 2. The GSM radio shows a mean power of 67.3 ± 20.9 mW, only 7.9 mW bigger than idle over the total length of the benchmark, and accounting for twenty-two you look after the mixture power (excluding backlight). All alternative elements showed associate degree RSD of below 3 %.

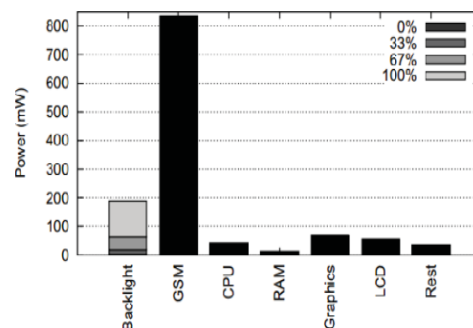


Fig 3.GSM phone call average power

Figure 3. Shows the ability consumption once creating a GSM telephone call of 57-second period [1]. GSM power clearly dominates during this case at a median power of 832.4 ± 99.0 mW. the ability breakdown for emailing between the GPRS and Wi-Fi benchmarks is comparable, apart from the GSM and Wi-Fi radios. Despite presenting identical workloads to the radios, GSM consumes over 3 times the ability of Wi-Fi. For net browsing application, GPRS consumes a lot of power than Wi-Fi by an element of two.5. just in case of a next-generation technology 3G, this issue might increase even a lot of thanks to the upper bandwidths supported by 3G. The results of this study

show that the bulk of power consumption are often attributed to the GSM module and also the show, as well as the alphanumeric display panel and touchscreen, the graphics accelerator/driver, and also the backlight.

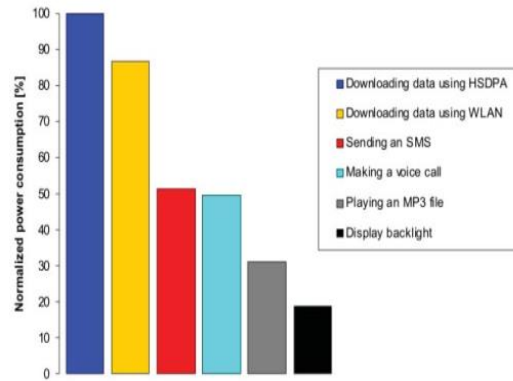


Figure 4 . Power consumption for different services

In Figure 4. Power consumption for various mobile services normalized facility to the consumption throughout downloading information victimisation High-Speed Downlink Packet Access (HSDPA) is shown [2]. Downloading information victimisationWireless native space Network (WLAN) relatively drains lesser power. For causing associate SMS and creating a voice decision, the facility consumption is sort of comparable however considerably but that of HSDPA and wireless local area network. It is clear from the plot that each one the capabilities that imply the employment of a wireless air interface area unit power- hungry, there for considerably reducing the battery lifetime of the phone. The paper reports larger energy consumption in 4G networks for text electronic communication and voice services than energy consumption in 3G networks. On the opposite aspect, the 4G networks become additional energy friendly once giant volumes of information got to be downloaded additionally 4G provides the fastest speed as compare to others that ne'er seen before.

Table – 1 Power consumption for voice service

scenario	GSM	UMTS	LTE,WIMAX
Receiving a voice call	612.7mW	1224.3mW	1838.1mW
Making a voice call	683.6mW	1265.7mW	2060.3mW
Idle mode	15.1mW	25.3mW	32.7mW

Table .1 shows the values for power consumption on Mobile for GSM, UMTS, and LTE mistreatment the voice service. The ability values for the calls are obtained by creating and receiving a call of 5 minutes and conniving the common of the ability levels. The observation results show that creating a decision mistreatment GSM-47%, UMTS-56%, and LTE-72%. For receiving a decision GSM-48%, UMTS-59%, LTE-74% and that i ideal mode GSM-15%, UMTS-25%, LTE-33%.LTE takes perpetually additional power as compared to others.

Service	2G	3G	4G
50 SMS of 100 bytes	90J	110J	150J
100 Mbytes downloading	10002.6J	3512.1J	2131.2J
5 Hours voice calls	12304.8J	22782.6J	28638.6J
Total	22404.0J	26404.7J	30919.8J

Table - 2 Energy comparisons using 2G, 3G, 4G

Table .2 shows a comparison of energy consumption mistreatment a pair ofG alone, 3G alone and intelligent shift [2] between these networks. it's clear from the observation that, 4G consumes additional power than 3G for the text of the appliance electronic messaging, downloading knowledge and voice decision. Parallel connections save energy however the gains vary betting on the technology. transmission control protocol downloads throughout 3G voice calls result into 75%-90% energy savings, transmission control protocol downloads throughout VoIP calls result into 30%-40% savings and transmission protocol control downloads once different transmission control protocol streams square measure active at identical interface result into 0%- 20% savings. The results indicate that there's a big potential to avoid wasting energy if applications square measure designed to require advantage of this development.

V.WIRELESS TECHNOLOGY EVOLUTION

The first generation (1G) mobile wireless communication network was analog used for voice calls solely. The second generation (2G) could be a digital technology and supports text electronic communication. When this was 3G that provided transmission support in conjunction with higher knowledge transmission rates and enlarged capability. The fourth generation (4G) integrates3G with fastened net to support wireless mobile net, that is an evolution to beat the constraints of 3G and conjointly raises the QoS, will increase the information measure and reduces the price of resources.

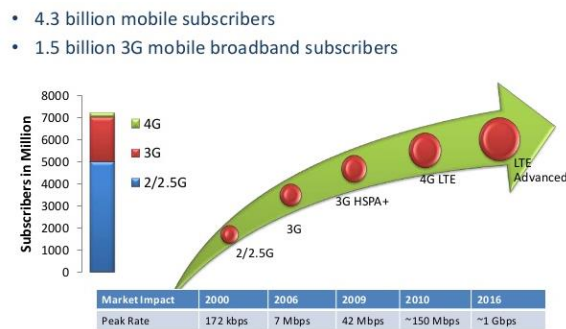


Figure 5 . Wireless Technology Evolution

Figure 5. shows the entire wireless technology evolution from the year of 2000 to 2016, however wireless technology bit by bit rising the techniques furthermore as speed (172 kbps to 1gbps) still improvement goes on and axis shows the scores of subscribers exploitation this technology .we used 2G for long length as compare to others at the moment 3G, 4G and these days we tend to area unit exploitation 4G LTE advance which supplies speed up to 1Gbps.

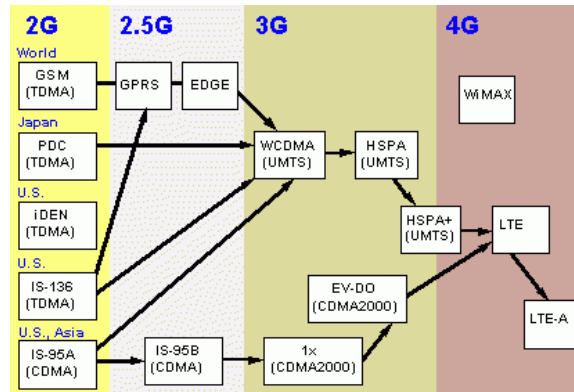


Figure 6. Evolution digital cellular standards architecture

Figure 6. Shows the evolution of digital cellular standards design in 2G we have a tendency to used GSM, in 2.5 G used GPRS and EDGE, in 3G used UMTS and in 4G we have a tendency to victimization LTE(wi-max) once following version of 4G is LTE advanced which supplies the most effective performance ever.

VI. CONCLUSIONS

This comparative survey given the Associate in analysis of power consumption and wireless technology evolution in smart phones technology in numerous use case state of affairs like causation an SMS, creating a voice decision, receiving a decision downloading information and plenty of alternative things. the end result of this analysis is that the facility consumption are a lot of and demanding in phone's electronic equipment and thus are often thought-about as an honest candidate for optimizing power consumption. The surveys say that within the feature we should always be a lot of specialize in power savings techniques.

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